LESSON 9 • TERM 2 • LEVEL 1

Picture Equations

A shape hides a number

TODAY WE WILL

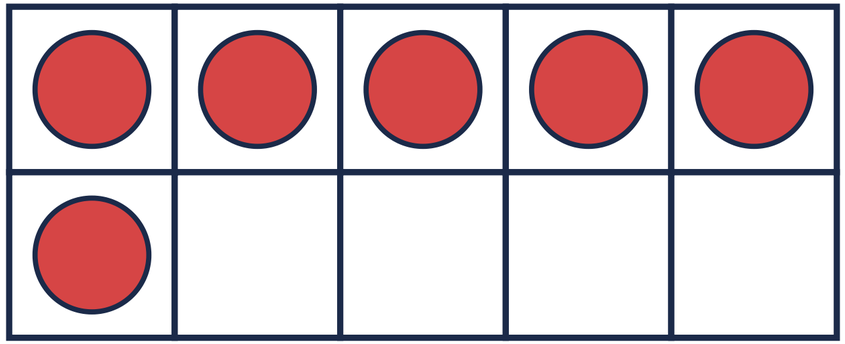
Find the number a shape is hiding.



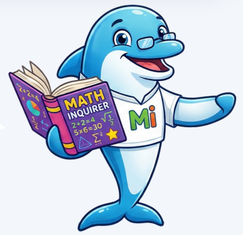
Warm-up: a box hides a number

2–8 MIN

Touch the counters. Say the number in the box.



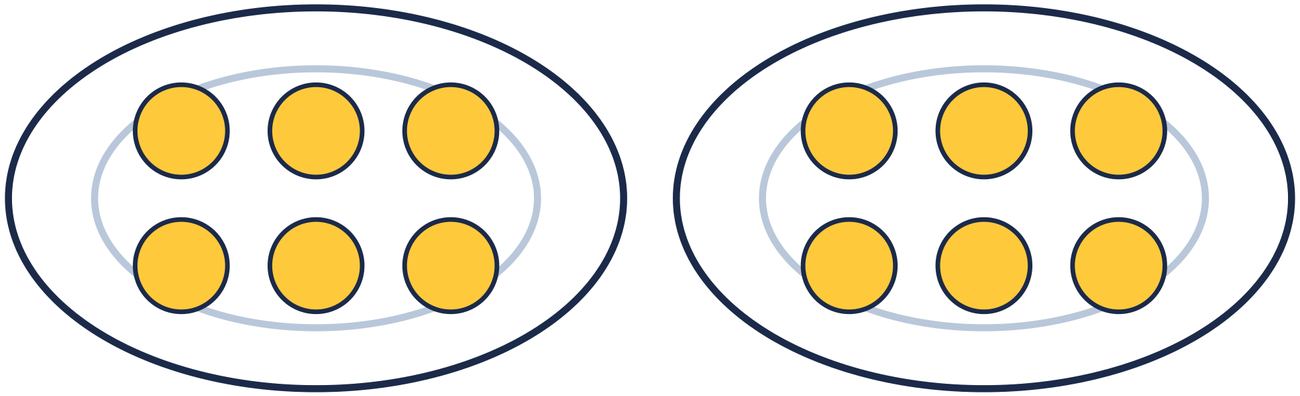
6	+	☐	=	10
☐	+	3	=	8
9	=	5	+	☐



A box can hide a number!

The same number twice

2–8 MIN

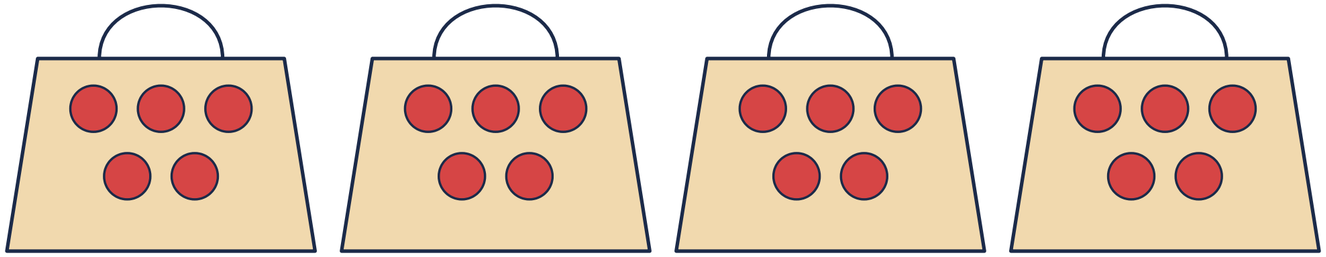




Two plates with the same number of counters.

4 groups of 5

2–8 MIN



4	×	5	=	

4 × 5 means 4 groups of 5.

Sneak-peek puzzles

8–12 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

PUZZLE 1: anyone can try

A cup hides some counters.

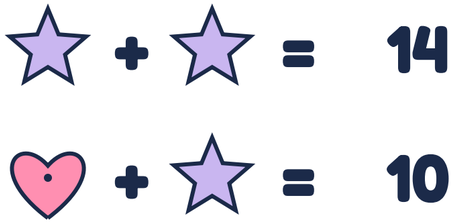
The cup and 4 loose counters make 9 in all.

How many counters are in the cup?

PUZZLE 2: think

Each shape is a number.

How many is the heart?



Explore: a shape and a pile

12–24 MIN

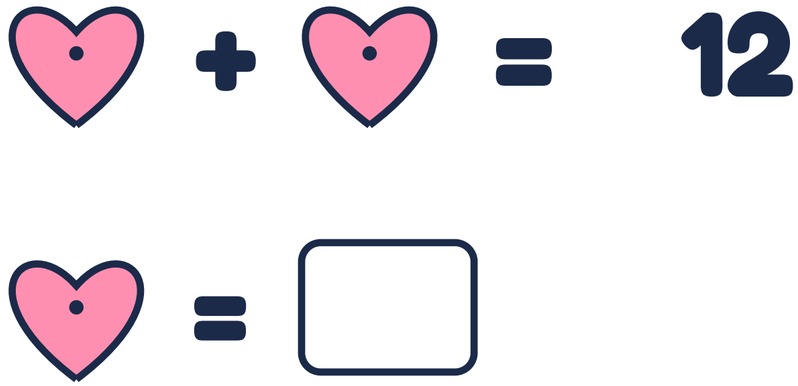




1. Count out 10 counters.
2. Put 4 on the square card.
3. The rest go on the circle card. How many?

The same shape twice

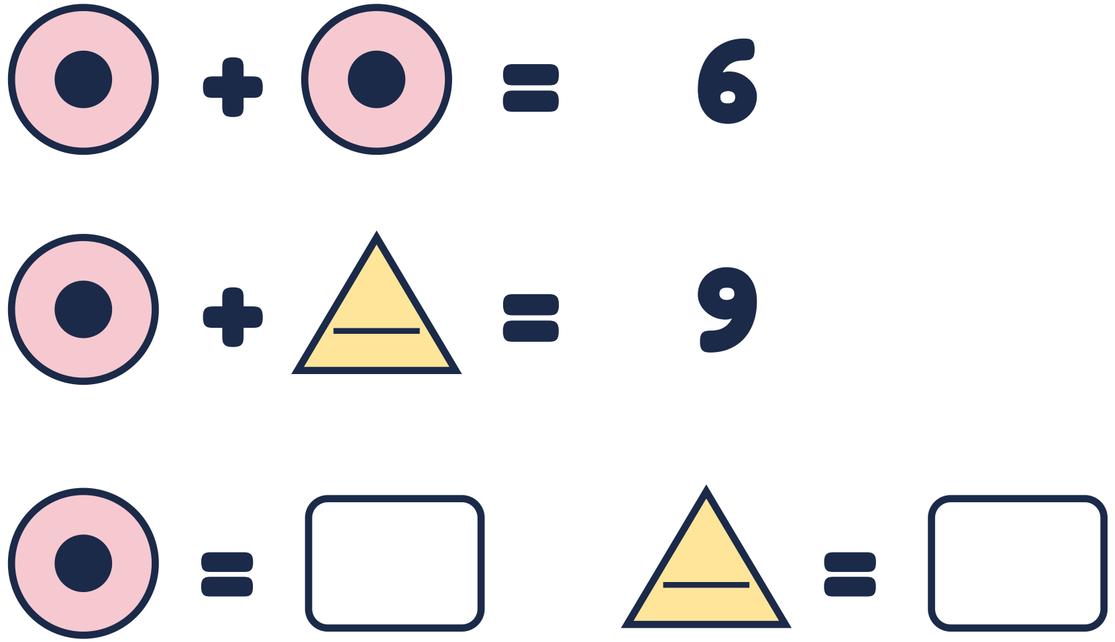
12–24 MIN



1. Count out 12 counters.
2. Share them onto 2 heart cards, one at a time.
3. The piles must match. How many on one heart?

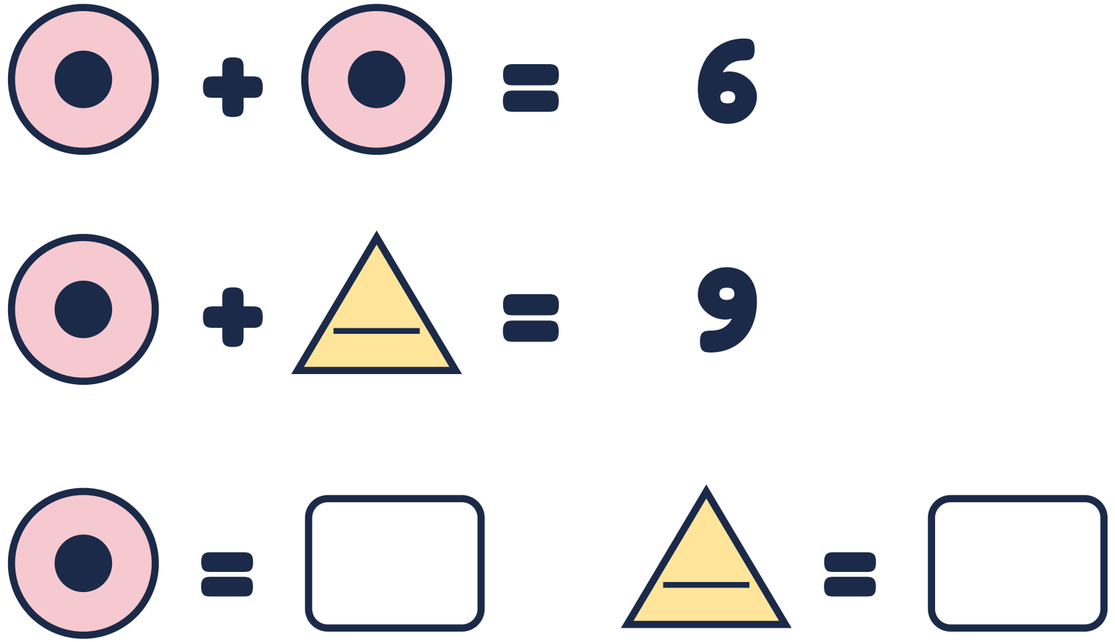
Two lines, same shapes

12–24 MIN



Two lines, same shapes

12–24 MIN



- Line 1 tells you the circle.
- The circle is the same number in line 2.

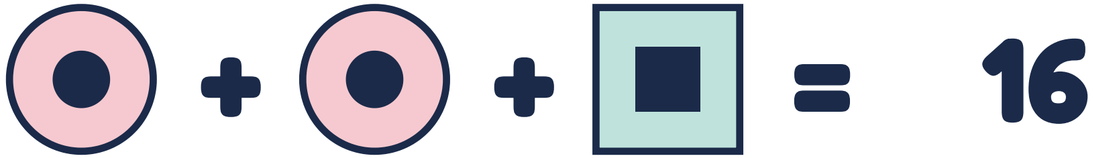
Three rules

24–36 MIN

1. Each shape is a number.
2. The same shape is the same number every time.
3. In our puzzles, different shapes are different numbers.

Find it, then check it

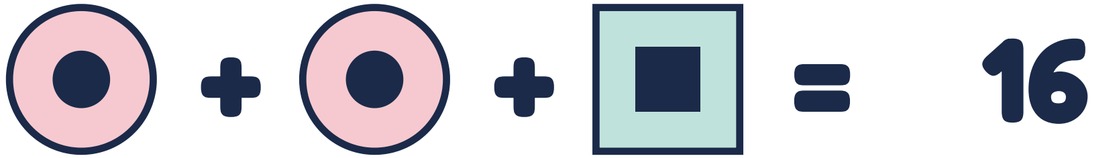
24–36 MIN





Find it, then check it

24–36 MIN





STEPS

Take the square away from the total.

Share what is left between the 2 equal circles.

Check: put the number back into the line.

Take one line from another

24–36 MIN





Take one line from another

24–36 MIN





STEPS

Line 1 has one more star than line 2.

The totals show how much that extra star is.

Then put the star into line 2 to find the triangle.

24–36 MIN

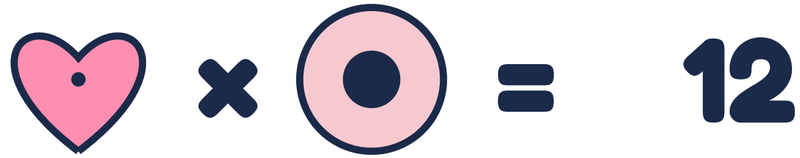
Check in both lines

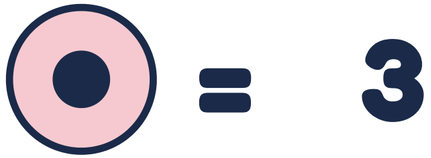
Line	Check	Does it match?
Line 1	5 + 5 + 7 = 17	Yes
Line 2	5 + 7 = 12	Yes

Check in every line, not just one.

A times line

24–36 MIN





STEPS

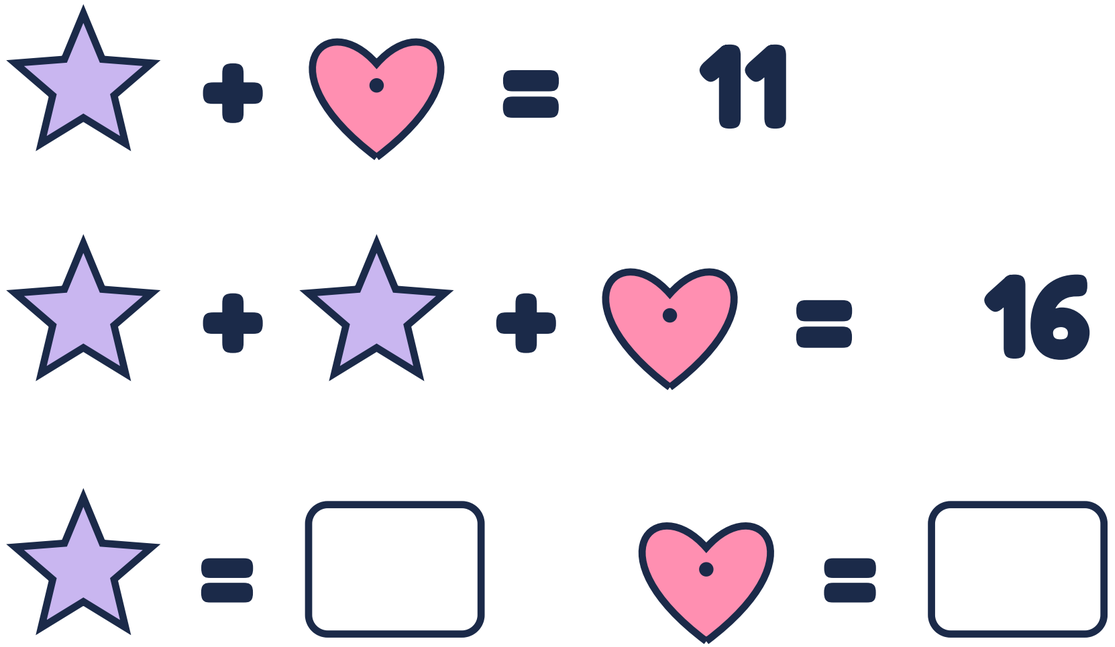
heart × circle means heart groups of the circle.

How many groups of the circle make the total?

Check: put the number back into the line.

Your turn

24–36 MIN



Find the star and the heart. Then check both lines.

Check: your turn

24–36 MIN

Shape	Number	Check
Star	5	Line 1: 5 + 6 = 11
Heart	6	Line 2: 5 + 5 + 6 = 16

36–40 MIN

Clap and whisper

Count to 20 with me. Clap on 5, 10, 15 and 20.

Whisper the numbers in between.

Stand up. Sip some water.

Workbook page 1: Start here

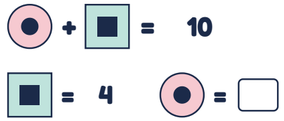
40–70 MIN

Questions 1 to 4. Take your time.

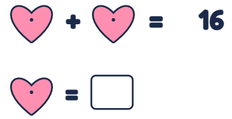
S1 A cup hides some counters. 3 counters are outside the cup. There are 11 counters in all. Fill in the blank.



S2 Each shape is a number. Find the number for the circle.



S3 Each shape is a number. Find the number for the heart.



S4 Each shape is a number. Find the number for the star.



Check page 1

40–70 MIN

S1	8
S2	6
S3	8
S4	5

Workbook page 2: Start here

40–70 MIN

Questions 5 to 7. Take your time.

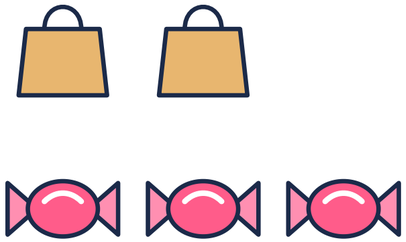
S5 Each shape is a number. The square is 4. Fill in the blanks.



S6 Ben says the circle is 7. Look at the picture.



S7 Each bag has the same number of sweets. 3 sweets are outside the bags. There are 13 sweets in all. Fill in the blanks.



Check page 2

40–70 MIN

S5	10; 5
S6	No; 6
S7	10; 5

40–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	Two lines and a times line
Go further (page 4)	Three shapes at once
Investigation (page 5)	One line or two?
Check	Finish and check your own work

Everyone may choose. Not finished? Pages 3 to 5 are for home too.

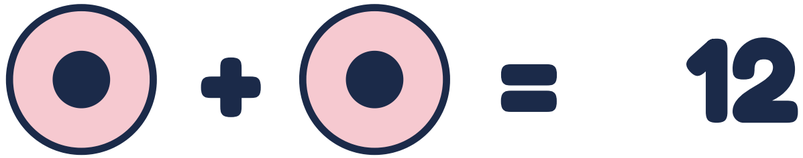
Talk and explain

70–76 MIN

AISHA SAYS

Circle + circle = 12.

So circle + circle + circle = 18.



Is Aisha right? How could we check?

I think … because …

70–76 MIN

Back to the sneak-peek



14



10



= 7



= 3

The star is 7. The heart is 3. The cup holds 5.

Exit ticket

76–82 MIN

On your own. No looking back.

E1 **Each shape is a number. The triangle + the triangle = 18. Find the number for the triangle.**

The triangle is ▢.

E2 **The heart + the square = 13. The square is 5. Find the number for the heart. Choose the correct answer. Write its number in the bracket.**

E3 **Each shape is a number. The star is 5. Find the number for each circle.**



Each circle is ▢.

Check the exit ticket

76–82 MIN

E1	9
E2	(4) 8
E3	7

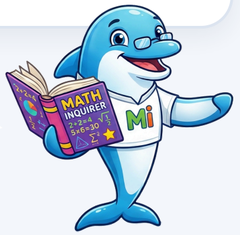
Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ finding the number for one shape ☐ using two lines ☐ checking in every line ☐ something else
What helped me	☐ counters and shape cards ☐ drawing it ☐ checking my answer ☐ I was not stuck
Next time I will try	☐ using counters ☐ checking my answer ☐ drawing it ☐ asking a question
	☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

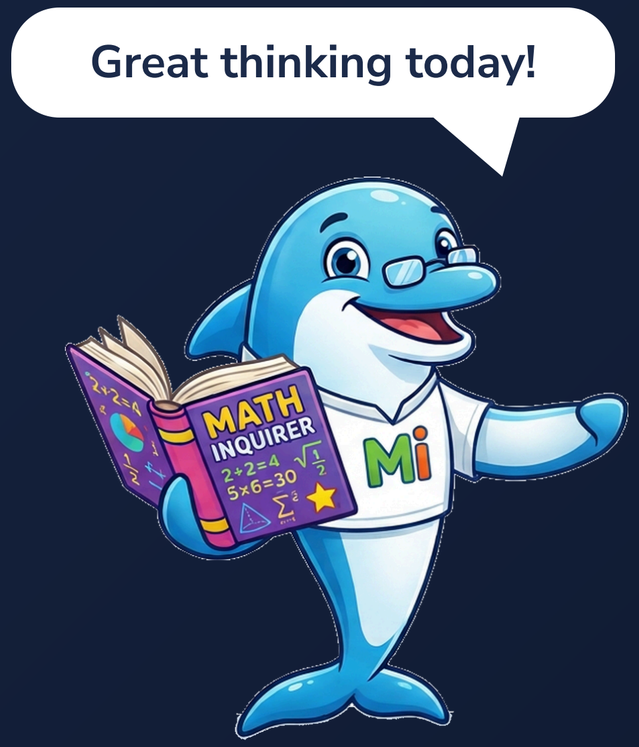
Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

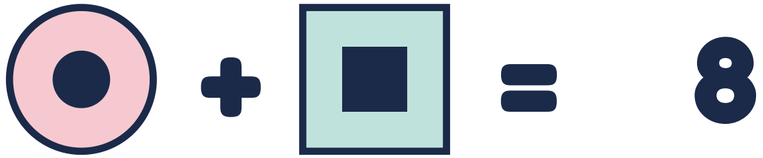
Find One First, and Mi's Challenge Quest

Have ready: the Term 2 Lesson 10 workbook with its tokens page cut out (30 tokens), 3 to 6 small things, the Quest Log in its envelope (keep it closed), a quest partner, a pencil and paper.



Go further: one line or two?

OPTIONAL



Could the circle be 2? Could it be 3?



What does the second line tell us?

OPTIONAL

If we have time: pick one

Activity	What we do
Can a shape change?	Ali says the heart is 5 in one line and 6 in another.
Make a puzzle	Pick two numbers. Write two lines for your grown-up.
Teach it back	Show a toy how a cup hides a number

Optional. Everyone may join.